

EDUCATION

- **Pusan National University (PNU)** Busan, Korea
M.S. in Electrical & Electronic Engineering Mar 2024 – Feb 2026
 - **Thesis:** Two Studies in Robotic Manipulation: Development of a Lightweight Compliant 6-DoF Arm and Dynamic Non-Prehensile Object Transfer
- **Pusan National University (PNU)** Busan, Korea
B.S. in Electrical Engineering Mar 2018 – Feb 2024
 - **Military Service:** Oct 2018 – Jun 2020

COMPETITIONS

- **RoboCup@Home 2025 (Salvador) – First Place:** Developed a lightweight compliant 6-DoF manipulator (HW & SW); contributed controller, motion planning, and high-level control API.
- **RoboCup@Home 2024 (Eindhoven) – Second Place:** Developed arm controller, motion planning modules, and high-level control API interface.
- **RoboCup@Home 2023 (Bordeaux) – First Place:** Hardware engineering contributions.
- **ZEUS Industrial Robot ZERO Mission Challenge 2024 – First Place (Presidential Award, Republic of Korea):** Team lead for “Robot Bartender”; implemented arm controller and motion planning; integrated an LLM-based interaction pipeline.
- **Space Exploration & Development Robot Challenge 2023 (Korea) – First Place (Prime Minister’s Award, Republic of Korea):** Developed a 4-DoF rover arm hardware and controller for a lunar-environment mock mission.
- **Baemin Food Delivery Challenge (Real Robot) 2023 – Second Place (President’s Award, Korea Institute for Robotics Industry Advancement; KIRIA):** Developed a low-level motor driver interface for the real-robot delivery platform.
- **Baemin Autonomous Delivery Challenge (Simulation) 2023 – First Place (Ministerial Award, Ministry of Trade, Industry and Energy; MOTIE):** Contributed to simulation task setup and overall performance.

EXPERIENCE

- **OMRON SINIC X** Japan
Incoming Research Intern Jul 2026 – TBD
 - **Research Topic:** Learning-based dynamic collaborative manipulation.
- **Tomorro Robotics** Korea
Research Intern Sep 2025 – Present
 - **CES 2026 Demo System (Autonomous Factory-Line Collaboration):** Lead engineer for the autonomous mobile manipulator in a dual-robot workflow. Implemented the complete pipeline for the mobile base to detect, grasp, and transport boxes to a conveyor, ensuring precise synchronization with a secondary VLA-based packing robot.

PUBLICATIONS

- **Dynamic Scoop-and-Flick Manipulation for Rapid Non-Prehensile High-Arc Object Transfer:** G. Ahn*, J. Lee*, S. Oh, M. Shin, S.-J. Yi, and J. Seo, *IEEE International Conference on Robotics and Automation (ICRA) 2026*.
- **ANUBIS: A Compact, Low-Cost, Compliant Humanoid Mobile Manipulation Robot:** T. Kang, J. Kim, S. Nasrat, D. Song, G. Ahn, M. Jo, S. Lee, S.-J. Yi, *IEEE-RAS Humanoids 2025*.
- **Lucio at RoboCup@Home: An Open-Hardware Mobile Manipulator with Modular Software and On-Device LLM Planning:** D. Song, T. Kang, S. Nasrat, J. Kim, M. Jo, G. Ahn, S. Lee, S. Yi, *IEEE-RAS Humanoids 2025* (Poster Presentation).
- **Development of Dual-Arm Human Companion Robots That Can Dance:** J. Kim, T. Kang, D. Song, G. Ahn, S.-J. Yi, *Sensors* 24(20), 6704.

PROJECTS

- **Autonomous Quadruped Patrol System (Gov./ National Police Agency Related):** Built perception coverage for safe navigation using four Intel RealSense D435 cameras to minimize blind spots. Owned camera mount design and multi-camera depth pipeline (stitching / fusion) to generate obstacle maps for collision-free locomotion while patrolling designated routes and detecting anomalies (suspects / illegal parking).
- **Mobile Dual-Arm Robot System for Scalable Learning Data Collection (LG Electronics Industry-Academia):** Developed a mobile bimanual manipulation platform optimized for data collection and safe policy rollouts. Led development of a lightweight compliant manipulator to reduce risk of environment damage while keeping the system cost-effective; later expanded into M.S. thesis direction.

SKILLSET

- **Technical Areas:** Kinematics & Dynamics, Sensor Fusion, Control, Motion Planning, Optimization, Object Recognition, Deep Learning
- **Languages:** C++, Python, Matlab
- **Frameworks:** ROS / ROS2, MuJoCo, Webots, Isaac Gym, Drake
- **Software Tools:** CAD(Onshape), Git, Docker, PyTorch, OpenCV, CMake, Make
- **Team-based Development:** Git-based collaboration (branching, code reviews) and collaborative CAD part design and assembly in Onshape for integrated robot hardware.